Responsible group work points

From Lecture 0:

The group gets an additional 5*N_group_members points that can be distributed among the group members based on contribution, reliability, and collaboration.

Contribution - How much and how well the student contributed to the actual group work?

Reliability - How well the student followed the agreed-upon tasks and schedule?

Collaboration - How did the student work with others?

This way, students who conduct responsible group work get rewarded.

Teachers have the right to change the distribution of these points if we see a discrepancy. Journaling and logging of activities would be used, together with physical meetings, if needed. If the student doesn't contribute to the group work sufficiently, teachers have the right to significantly reduce the student's final grade or fail the student completely.

Exploratory and Sentiment Analysis of Yelp Reviews using Apache Spark — Group 12

Member 1: Arshman Tariq — Points: 5

Justification:

Contribution: Arshman was solely responsible for the complete EDA notebook, which represents one of the two major analytical deliverables of the project. This includes all data exploration across five tables, the custom business category taxonomy (mapping 200+ tags to 10 macro-groups), all geographic visualisations (Basemap and Folium), the star rating distribution, review length analysis, temporal trend analyses, word frequency analyses, category-level sentiment breakdown, and the open/closed business status analysis. She also co-authored the report and presentation with equal involvement.

Reliability: All exercises were submitted on their respective deadlines. The preliminary report draft was submitted on 6 April. Arshman consistently followed through on her agreed responsibilities — the EDA notebook was completed on schedule, allowing her to improve her EDA findings and interpretations while Faizan finalised the ML pipeline, which shows reliable self-management and team awareness.

Collaboration: Being roommates facilitated continuous, natural communication. Arshman actively engaged with Faizan's ML findings and contributed to framing Business Reputation Analysis as a joint output from both tracks. She reviewed and gave feedback on the ML notebook's feature selection and model interpretation sections, and her EDA findings directly informed feature engineering choices in the ML pipeline.

Member 2: Muhammad Faizan Tanveer — Points: 5

Justification:

Contribution: Faizan was solely responsible for the complete ML/Sentiment Analysis notebook, which represents the second major analytical deliverable. This includes the multi-table join, all feature engineering (15 engineered features), the full ML pipeline (TF-IDF, CountVectorizer, StandardScaler,

VectorAssembler), class imbalance analysis and stratified undersampling, training and evaluation of three classifiers (LR, RF, MLP), the correlation matrix analysis, LR coefficient-based feature importance, and the Business Reputation Scoring module (scoring 6.99M reviews, temporal trend analysis, state-level analysis, and open-vs-closed sentiment comparison). He also co-authored the report and presentation with equal involvement.

Reliability: All exercises and the preliminary report draft were submitted on their respective deadlines. Faizan remained committed to quality over speed — when model training on Rahti encountered memory and time constraints, he investigated and resolved the technical limitations (MLP solver switch, dataset sampling strategy) rather than settling for a broken result. This reflects reliable, self-directed problem-solving.

Collaboration: Faizan engaged actively with Arshman's EDA outputs, particularly the category taxonomy and star distribution findings, which directly motivated the class balancing strategy and label definition in the ML pipeline. He initiated the discussion that led to the Business Reputation Analysis — recognising that standalone sentiment classification was insufficient for a meaningful contribution — which became the project's most novel analytical result.

Additional sources for assessment:

- Both individual Jupyter notebooks are available in the shared Rahti workspace and can be assessed to verify independent, parallel development and equal volume of work.
- The activity journal is the primary log of weekly contributions.
- The contribution section of the submitted project report confirms the role split and joint report writing.
- Rahti session logs will show separate login activity from two different devices, confirming independent development.
- All the material is also uploaded to the GitHub Repository, which includes project files and outputs of both notebooks.

  GitHub - FaizanTanwir/BDPA2026_Group-12_Project-YELP · GitHub

- Some of the outputs which were excluded from the report are also there, confirming the collaborative and equal volume of work.

Any additional sources we should consider when doing assessments? (activity logs, etc) Please add the links, paste images, etc., if needed.